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Thalassogigantia insuliformis

A Biological Monograph on the Tanayu

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Notice

The present monograph was compiled almost two millennia prior to the withdrawal of Senate authority from AR-29G and reflects observations conducted while Tanayu husbandry practices were alive as part of humanity's vast knowledge on ecosystem management. However, recent covert missions to AR-29G have failed to confirm that the content herein accurately reflects contemporary conditions.

Assessment

Current inhabitants of AR-29G do not appear to possess the knowledge or skill necessary for the deliberate redistribution of Tanayu offspring and Tanayu archipelago management, all of which are described in this volume. Readers are advised accordingly that the section on *Relations with Human Civilizations* should be interpreted as accounts of documented historical practice.

This assessment should not be considered definitive, as no inhabitant of AR-29G was questioned due to post-withdrawal non-interference protocols. It is instead derived from the observations conducted during long-term covert missions and from the apparent absence of such practices in contemporary AR-29G societies.

Authorized by Directive 14-88,
Office of Abandoned Colony Reassessment

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Senior Colonial Assessor

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1 Introduction

Commonly known as the Tanayu, *Thalassogigantia insuliformis* is one of the most impressive recorded examples of pelagic host megafauna. Mature specimens carry complex ecosystems on their backs, resembling mobile islands of white soil and red vegetation, and sustain permanent human settlement.

[redacted]

Figure 1: Depiction of the Tanayu-borne city-state of [redacted] as seen from afar.

Although *T. insuliformis* has long been observed, systematic xenobiological study remains difficult. Its extreme longevity, slow movement, and broad geographical distribution mean that many of its most important life processes occur over timescales much longer than those of human researchers.

The present document summarizes the current biological understanding of *T. insuliformis*, with particular attention to its taxonomy, external morphology, carapace physiology, life cycle, symbiotic relations, and ecological role as a host substrate for human civilizations.

2 Nomenclature

The Kupina Hala term *Tanayu* is the preferred Interstellar Xenobiology Bureau common name. In Kupina Hala, it is traditionally analyzed as *tana* ”land” and *layu* ”to live,” giving an approximate sense of ”living land.” Some, however, maintain that the name derives instead from *Tanen Yu*, a historical figure associated with the earliest recorded settlements on AR-29G.

Due to the lack of sources confirming or denying either claim, the etymology of *Tanayu* remains disputed to this day.

3 Taxonomic Classification

Rank	Classification
Family	Thalassogigantidae
Genus	*Thalassogigantia*
Species	*Thalassogigantia insuliformis*

Table 1: Accepted taxonomic placement of the Tanayu.

The genus *Thalassogigantia* contains three commonly cited species. The best attested is *T. insuliformis*, the true Tanayu. [redacted]

Species	Common Name	Status
T. insuliformis	Tanayu	Extant
[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]

Table 2: Recognized and reputed members of *Thalassogigantia*.

4 General Morphology

Mature specimens of *T. insuliformis* are massive marine organisms that inhabit the ocean surface on AR-29G. They remain mostly submerged, but their backs protrude above the waterline, supporting a calcareous carapace that acts as the foundation for entire ecosystems. In large adults, this exposed dorsal region can expand several kilometers in diameter, with some specimens approaching [redacted] kilometers.

The submerged anatomy of *T. insuliformis* is loosely reminiscent of turles, giant manta rays, and horseshoe crabs. It swims slowly and deliberately using lateral oar-shaped limbs and undulations of a tail that constitutes [redacted]% of its body length.

From a human's perspective, *T. insuliformis* moves at a speed akin to that of a drifting vessel, rather than that of other marine animals. Tanayu-borne settlements experience their host's migration as a gradual change of climate, currents, and visible stars rather than perceptible motion.

Figure 2: Illustration of a *T. insuliformis* speciment as seen from above, showing its long tale and swimming extremities.

T. insuliformis possesses an exposed, baleen-like filter feeding structure that encompasses the entirety of its anterior surface, beginning above sea level and reaching hundreds of meters below the waterline. This structure passes enormous quantities of water at a constant rate, extracting plankton-like organisms and, given the size of mature specimens, anything else in their way.

Figure 3: Anterior feeding structure of mature *T. insuliformis*. Left: external view of the baleen-like plates. Right: Image of a ship behind filtered through into the animal's mouth.

Filter-feeding supplies part of the energy needs of *T. insuliformis*, but it does not sufficiently meet the metabolic requirements of mature adults. The remaining energy is supplied through extensive symbiotic relationships with photosynthetic organisms living on the carapace.

5 The Carapace

5.1 Composition

The most impressive anatomical feature of *T. insuliformis* is its carapace, a massive structure formed from a calcareous secretion called *albasoil*. This secretion is released through various pores upon the animal's dorsal region and contains waste products from its metabolism, making it abundant in nutrients. In this sense, the carapace may be understood as a monumental biological waste-processing system.

Albasoil hardens rapidly after secretion. Continued secretion beneath older deposits gradually displaces the hardened material upwards, which progressively grows the carapace. Upper layers are exposed to wind, rainfall, biological abrasion, and thermal stress, which eventually erodes the albasoil into sand-like particles.

[redacted]

Figure 4: Depiction of the different carapace layers in a mature *T. insuliformis*. Upper layers are eroded and sand-like, while lower layers are hardened and unaffected by the elements. Several topographic features like basins and ridges can be observed.

This erosion results in ridges, basins, fissures, terraces, and elevated spines on the carapaces of mature specimens. Human settlement patterns are strongly shaped by this topography. Stable basins collect water and albasoil, which is the basis for agriculture in AR-29G. Conversely, elevated ridges can be used for observation and defense.

5.2 Regulation

Field observations suggest that *T. insuliformis* can influence where new albasoil is deposited, rather than secreting uniformly across the entire carapace. Individuals appear capable of concentrating deposition in regions where erosion has exposed older layers or where ecological productivity has declined.

When dorsal secretion is unnecessary or potentially harmful, *T. insuliformis* can excrete through lateral pores submerged underwater, discharging albasoil into the surrounding sea. Lateral secretion prevents excessive carapace growth and distributes nutrients to nearby marine ecosystems. For most host civilizations, the process is known only indirectly through the increased biological activity observed in the waters surrounding their host.

Moreover, *T. insuliformis* does not produce albasoil at a constant rate throughout life. As individuals reach a certain size and age, their secretion rates appear to diminish, adding a second safety measure to prevent uncontrolled accumulation of albasoil.

Figure 5: Observed dorsal secretion patterns in a mature individual over a three-century interval, with quantities measured every twenty years.

The mechanism governing this regulation remains incompletely understood. It is likely to involve both internal metabolic signals and external feedback from the carapace ecosystem. A heavily vegetated, nutrient-stable area may require less new secretion than a damaged or recently eroded one.

6 Symbiotic Ecosystem

T. insuliformis has developed an indispensable symbiotic relation with tadpole algae. These organisms, reminiscent of plants and trees, attach to the host during its juvenile state, when it is completely submerged underwater, and lie dormant until the carapace surfaces. They then expand like clonal groves.

The albasoil covering the carapace acts as as a natural fertilizer for the algae. In exchange, they produce sugars and other organic compounds using sunlight, water, and atmospheric carbon dioxide, and provide a portion of the generated energy to the host through an intricate network of roots.

This solar energy plays a key role in the nutrition of mature *T. insuliformis*. In times of food scarcity, individuals will stop swimming and let themselves drift with the currents, recharging their energy exclusively through this symbiotic relationship—and through anything that may inadvertently wander into their mouth.

Figure 6: Depiction of albasoil generation and algae root conduits forming the basis of the symbiotic system on the carapace.

7 Life Cycle

7.1 Reproductive Gatherings

All specimens of *T. insuliformis* converge every few centuries at a specialized breeding ground to reproduce, creating the appearance of a continent when their carapaces stand side by side. These immense gatherings represent one of the most significant biological events on AR-29G and are central not only to Tanayu ecology but also to the political life of many host civilizations.

T. insuliformis is a simultaneous hermaphrodite species, and during these gatherings each individual both lays eggs and fertilizes those of others. The eggs hatch after only a few weeks.

7.2 The Tanalet Stage

Newly hatched individuals of *T. insuliformis*, known as *tanalets*, are approximately the size of a human hand. Soon after emerging, they instinctively identify the nearest individual as their mother, regardless of genetic parentage, and will remain beneath or immediately behind them, sheltering within the hydrodynamic wake produced by the larger animal.

It is during this state that the carapace of tanalets is colonized by tadpole algae. However, they will remain dormant while the host remains submerged, so tanalets depend entirely on filter feeding for energy. They consume plankton-like organisms in surface waters and frequently compete with siblings for food resources left behind by the adult they follow.

Predation, starvation, and competition lead to extremely high mortality during this stage. Of the thousands of tanalets following a specific adult, typically only three to five survive long enough to reach the next stage. Human intervention, however, can often increase survival rates substantially.

Figure 7: Maternal associations in early developmental stages. Left: newly attached tanalets occupying the hydrodynamic shelter zone beneath an adult. Right: mixed-age nursery aggregation at a reproductive gathering.

7.3 The Tanaling Stage

After approximately a hundred years, a surviving tanalet has grown big enough to rise to the surface and permanently expose its carapace above the waterline. Individuals in this developmental phase are known as *tanalings*.

The emergence of the carapace triggers the expansion of dormant tadpole algae, which gives tanalings access to photosynthetic energy for the first time. Their increased size also allows a broader diet. In addition to filtering plankton-like organisms, tanalings are capable of consuming fish and other comparatively large marine animals.

Socially, tanalings continue to accompany the adult they have followed since hatching. They behave similarly to adults in most respects, but remain socially subordinate because they have not yet acquired offspring of their own.

Tanalings commonly travel behind their mother alongside numerous siblings in a V-shaped arrangement—called *archipelagoes*—which allows each individual to exploit a different portion of the surrounding ocean while still maintaining cohesion with the group.

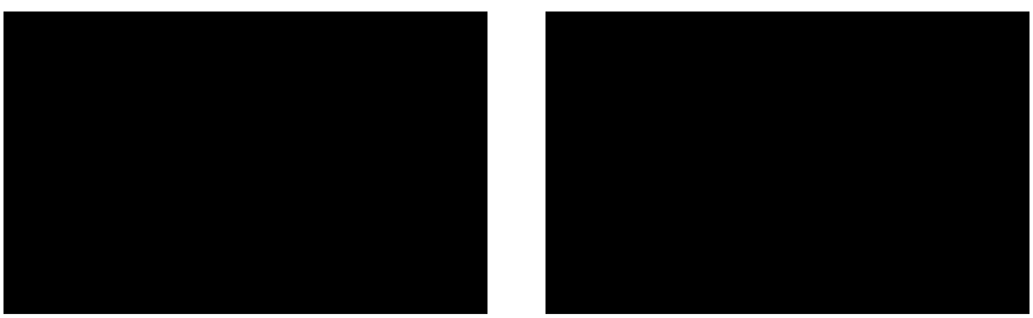

Figure 8: Representative tanaling behavior. Left: recently surfaced tanaling exhibiting early dorsal colonization by tadpole algae. Right: V-shaped formation during migration.

7.4 Transition to Adulthood

A tanaling that successfully acquires one or more tanalets during a reproductive gathering enters adulthood and leaves its mother's group to establish an independent migratory route. This departure naturally limits the long-term size of archipelagos in unmanaged populations.

Individuals that fail to acquire tanalets remain with their mother's group and await another opportunity at the next reproductive gathering.

7.5 Death

T. insuliformis can survive for millennia. Upon death, the body may remain afloat for a few years before eventually sinking into the ocean, taking the entire carapace ecosystem with it. If the deceased individual was leading an archipelago, succession depends on its composition.

If the adult had no tanalets under its care and the archipelago consisted solely of mature individuals, the oldest tanaling ordinarily assumes the role of principal adult, with younger siblings continuing to follow it until future reproductive gatherings allow them to acquire offspring of their own.

With dependent tanalets present, however, succession becomes less deterministic. The young disperse and seek protection beneath one of the accompanying tanalings. Any tanaling that acquires offspring in this manner immediately enters adulthood and departs to establish an independent migratory route, often accompanied by siblings that failed to obtain offspring of their own.

Tanalets that fail to find a tanaling—either because they become separated from the group or because no tanalings are present—have little chance of survival.

Figure 9: Observed terminal sequence in an elderly *T. insuliformis*.

8 [redacted]

[redacted]

9 Relations with Human Civilizations

9.1 Archipelago Management and Breeding Politics

As explained above, individuals of *T. insuliformis* naturally organize themselves into familial groups known as archipelagos, with a mature adult determining the migratory route.

In unmanaged populations, archipelagos tend to fragment over time as tanalings acquire offspring and leave their mother's group. This limits the long-term size and persistence of naturally occurring archipelagos. Many human societies, however, have altered this process by deliberately redistributing offspring among individuals under their stewardship.

Following a reproductive gathering, newly acquired tanalets may be transferred to an established adult, preventing tanalings that had acquired offspring from transitioning to adulthood and departing to found new archipelagos.

Similar interventions may occur following the death of a principal adult, when all dependent tanalets are placed under the care of a single tanaling in order to preserve the unity of the archipelago.

[redacted]

Figure 10: Depiction of tanalets being palced under the care of [redacted] following the death of [redacted], former principal adult of the [redacted] archipelago.

Human societies may also increase the size of their archipelagos by acquiring additional tanalets during reproductive gatherings. This frequently leads to large-scale conflicts between Tanayu-borne civilizations, which engage in underwater warfare to seize tanalets from one another and place them under the care of their own adults.

As a consequence, many of the great archipelagos inhabited today are not entirely natural social formations, but partially anthropogenic constructs maintained through centuries of careful intervention.

9.2 Human Ecology on the Carapace

Human diets on the carapace consist primarily of marine animals obtained through fishing and tadpole algae cultivated in albasoil. Excessive harvesting of these symbionts can weaken the host, so human societies have developed elaborate systems for managing the resources of the carapace. Rotational harvest systems, protected groves, and seasonal restrictions are widespread. In well-managed regions, human activity may even increase local biodiversity through the creation of ponds, terraces, compost fields, and managed woodlands.

Figure 11: Representative food production systems on a mature *T. insuliformis*. Left: cultivated albasoil terraces supporting mixed agriculture. Right: fishing vessels returning to the carapace with large catches of marine animals.

Human societies also have a vast knowledge of their archipelago's migratory route. Generations of observation allow communities to anticipate changes in marine resources and climate encountered along these routes. This knowledge forms the basis of agricultural calendars, navigation, trade, and religious practice.

10 Conclusion and Research Limitations

Current knowledge of *Thalassogigantia insuliformis* remains constrained by several factors. The species' extraordinary longevity makes direct observation of complete life cycles the work of several human generations. In addition, some of the most important physiological structures lie beneath kilometers of living tissue and calcareous material, rendering invasive study both ethically and practically unacceptable.

The extensive influence of human civilizations presents an additional challenge. Human activity has shaped many carapaces for centuries or millennia, making it difficult to distinguish unmanaged biological processes from landscapes and social systems produced through long-term stewardship.

Future research should therefore proceed in close cooperation with host civilizations and with due consideration for the exceptional ecological and cultural significance of the species.

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